PROJECT DETAIL · VERSION 8

AI-Powered Learning Assistant

A private enterprise learning platform for administrators, instructors, and students.

Platform

Web + Mobile

Features

8 modules

INTRODUCTION

Project overview

An intelligent, context-aware learning companion designed to personalize education for students and streamline workflows for educators.

Built directly into the coursework experience, the platform delivers personalized interactive tutoring, automated concept summarization, and smart knowledge querying. It bridges curriculum materials with real-time AI guidance to keep students engaged and instructors informed.

Administrators

Govern user operations

Instructors

Manage curricula and evaluations

Students

Engage in authorized coursework

INTRODUCTION

Eight features

01 **Authentication**

02 **Profile**

03 **Admin Dashboard**

04 **Classroom & Learning Material Management**

05 **Workspaces (personal & course)**

06 **AI-Generated Learning Resources**

07 **AI Tutor Chat & Summarization**

08 **Mobile App**

INTRODUCTION

Recent updates

Personal workspace	No longer linked to a classroom. It holds only the sources its owner adds.
Classroom structure	A classroom holds one or more courses.
Course workspace	New. Created inside a course; its sources are the course materials.
Course workspace sources	Students cannot upload or remove documents, because the instructor has already uploaded them.
AI tutor scope	Set by the workspace: a personal workspace uses my sources, a course workspace uses the course materials.

FEATURE 01 OF 08

Authentication

Login and forgot password

01

FOR INSTRUCTOR / STUDENT

Sign-in and password recovery

Sign-in

- Email or username + password
- First login requires a password reset
- “Remember me” option

Forgot password

- Input email
- Verify the OTP code sent by email
- Set and confirm a new password
- Redirect to the login page

Reset OTP and confirmation

Reset OTP

- Emailed as a single-use, time-limited code
- Valid for 2–5 minutes
- Invalidated after use or after a new request

After reset

- Confirmation email: “Your password was changed”
- Redirect to sign-in

FEATURE 02 OF 08

Profile

Instructor profile and student profile

02

FOR INSTRUCTOR / STUDENT

Instructor and student profiles

Instructor profile

- View profile
- Update profile
- Change password
- Organization (course, class)
- Contact: official email shown to enrolled students and admin only
- Logout

Student profile

- View profile
- Student ID
- Prompted after first login to add class, course (optional)
- Update profile
- Change password
- Logout

FEATURE 03 OF 08

Admin Dashboard

Users, classrooms and account security

03

FOR ADMIN

Managing users and View classrooms

Manage instructors

- **Create, update, delete**
  Add instructor accounts, edit their details, and remove accounts no longer needed.

View classrooms

- **View classroom members and courses**
  See every classroom, its instructors and students, and the names of its courses. Course files and resources stay private to the classroom.

Manage students

- **Create, update, delete**
  Add student accounts one by one, edit their details, or remove them.
- **Import Excel user list (students only)**
  Students log in with the emailed credentials, then are prompted to enter additional information.

KPI Card

- **Show Total KPI Card**
  All users list, total instructors, total students, total courses and students joined.

Account and security controls

User accounts

Edit, disable and enable users

Roles

Change roles (Instructor, Student)

Passwords

Trigger a password reset or force a password change

Audit log (Optional)

View login events, failed logins and active sessions

MFA

Require MFA setup for specific users

Password expiration and rotation policy

Admins can enforce automatic expiration intervals (e.g. 90 days) via Keycloak’s realm password policy.

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Classroom & Learning Material Management

Classrooms, courses, shared files and course workspaces

04

FOR INSTRUCTOR / STUDENT

Course structure and uploads

Instructor

- Create a classroom with one or more courses
- Invite co-instructors and students by link or code
- Manage students (remove from classroom)
- Upload files to a course: PDF, TXT, video, audio, PPTX, images, links
- Remove files already uploaded
- Join a classroom via code or link (co-instructor only)

Student

- Join a classroom by opening the instructor's invitation link
- Or join by typing the invitation code
- Browse all files and resources the instructor shared in each course
- Search and filter classroom files and resources by category or course name
- Create a course workspace to study a course's materials

Classrooms, courses and course workspaces

01 · CLASSROOM

Classroom

Created by the instructor. Students and co-instructors join by invitation link or code.

02 · COURSE

One or more courses

Each course is a subject inside the classroom and holds the learning materials the instructor uploads.

03 · COURSE WORKSPACE

Course workspace

Created inside a course. Its sources are the course materials, so students study without uploading other documents.

Why students cannot upload The instructor has already uploaded the course documents. Keeping course workspaces to those materials keeps AI answers, quizzes and flashcards aligned with the course.

FEATURE 05 OF 08

Workspaces

Personal and course workspaces for sources, chats, notes and study materials

05

FOR INSTRUCTOR / STUDENT

Workspace management

Instructors and students can

- **Create, rename and delete workspaces**
- **Add a description and optional cover or icon**
- **View and manage their workspaces**
- **Mark workspaces as favorites**
- **Add documents or sources (personal workspace)**

Two kinds of workspace

- **Personal workspace**
  Created from My Workspace. It holds the sources its owner adds and is not linked to a classroom.
- **Course workspace**
  Created inside a course. Its sources are the instructor's course materials.

Personal and course workspaces

	PERSONAL WORKSPACE	COURSE WORKSPACE
Where it is created	My Workspace page	Inside a course
Sources	Sources the owner adds	The instructor's course materials
Add or remove sources	Yes	No for students; the instructor manages course materials
AI tutor answers from	The owner's sources	The course materials
Notes, AI chats, resources	Yes, private	Yes, private to each student

What a workspace holds

Sources	Documents or sources in the workspace
AI chats	Conversations with the AI tutor
Notes	Personal notes
AI-generated learning resources	Flashcards, quizzes, multiple-choice, True/False, matching and fill-in-the-blank questions

Instructor and student personal workspaces

Instructor workspace

- **Prepare lessons**
  Collect sources, write notes, and draft material before uploading it to a course.
- **Chat with AI**
  Ask questions about their own sources and get answers with citations.
- **Generate with AI**
  Create quizzes and flashcards from their own sources.

Student workspace

- **Study on their own**
  Add personal sources and keep notes in a private notebook.
- **Practice with AI**
  Generate flashcards and quizzes for self-assessment.
- **Ask the AI tutor**
  Get answers grounded in their own sources, with citations.
- **Stay organized**
  Keep AI chats, notes and generated resources together in one place.

Course workspace

What students and instructors can do

- **Ask the AI tutor**
  Get answers grounded in the course materials, with citations.
- **Practice with AI**
  Generate flashcards and quizzes from selected course materials.
- **Keep notes and chats**
  Write notes and keep AI chats about the course in one place.

Instructors can also

- **Shared Resources(Quiz and Flashcard)**
  Share quizzes and flashcard sets generated from course materials with the course, so students can practice with them.

Sources come from the course

- **Course materials only**
  The instructor's uploads are the sources. Students cannot upload or remove documents.
- **Always up to date**
  New material the instructor adds to the course appears in the workspace.
- **Private study**
  Each student's notes, chats and generated resources stay private.

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AI-Generated Learning Resources

06

Generate quizzes and flashcards from workspace sources to help students study, review and practice.

FOR INSTRUCTOR / STUDENT

Difficulty levels and configuration

Before generating

- **Quantity**
  Choose how many questions or flashcards to create.
- **Difficulty**
  Pick Easy, Medium, Hard or Mixed.
- **Custom instructions**
  Optional. Name the topics, keywords or concepts to focus on.
- **Source materials**
  Choose which workspace sources to generate from: your own sources, or the course materials in a course workspace.

After submitting

- **Auto-grading**
  Submitted answers are evaluated automatically.
- **Final score**
  The student sees their final score when grading finishes.
- **Answer feedback**
  Show the correct answers with detailed explanations.

After generating

- **Review generated questions**
  Review the quiz or flashcards before starting. If needed, regenerate with updated settings.

Flashcards

Front and navigation

- Generate interactive study flashcards from selected learning materials
- Questions, technical concepts or terms on the front face
- Navigate with Next and Previous, or shuffle randomly

After flipping

- Detailed answers with source references
- Definitions
- Code syntax
- Review the referenced source material to reinforce learning

Four question types

Multiple choice (MCQ)

- **Single-select and multi-select**
  Students pick one answer, or several when more than one is correct.
- **Explanations with source references**
  Shows why each answer is right or wrong, with a link to the relevant source.

True / False

- **Practice statements**
  Students decide whether each statement is true or false.
- **Contextual feedback & source links**
  Explains why statements are true or false and references the original course slide.

Matching

- **Matching exercises**
  Pair items from two lists.
- **Concepts and definitions, source-grounded**
  Shows the correct pairs and why, with a link to the lecture slide for each.

Fill in the blank

- **Complete the sentence**
  Students type the missing word or phrase.
- **Answers with explanations**
  Displays the correct answer, rationale, and source reference to help students review mistakes.

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AI Tutor Chat & Summarization

Grounded answers, explanations and summaries

07

FOR STUDENT / INSTRUCTOR

Grounded answers with citations

Ask questions	About the sources in the current workspace: your own sources in a personal workspace, the course materials in a course workspace
Grounded answers	Based only on that workspace's sources
Hover Peek	Quick preview of a citation before you click
Contextual Follow-up Suggestions	Dynamically generates 3 relevant follow-up questions based on the current discussion to guide deeper exploration

Explaining in the student’s terms

Explaining clearly

- **Plain language**
  Difficult concepts broken down in plain words, step by step.
- **Examples and analogies**
  Ask for an example or an analogy. The tutor notes where an analogy stops being accurate.
- **Compare two concepts**
  Ask “X vs Y” for a table of similarities, differences and when to use each, with citations.

Beyond the materials

- **Web search fallback**
  If the workspace sources lack the answer, the tutor can search the web. Web results are labeled.
- **Their own language**
  Ask in Khmer or English and get the answer in the same language.

Summaries for documents and lectures

Documents

- Short summary
- Detailed summary: structured, sectioned, longer-form
- Key concepts: main ideas, each with a one-line explanation
- Important definitions
- Learning objectives: what the student should be able to do after studying
- Chapter-by-chapter summaries: modular, section-by-section breakdowns for lengthy readings and textbook chapters

Lecture video, link and audio

- Summarize a lecture video, link or audio file
- Each point links to a timestamp, so students can jump to that moment
- Key concepts and definitions show when they were mentioned

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Mobile App

Classrooms, courses, workspaces and the AI tutor on the phone

08

FOR INSTRUCTOR / STUDENT

Instructor on mobile: access

Login	Email or username and password
Browsing	Classroom → course
Search	Classrooms and courses
Viewing	PDF viewer that jumps to a page (PPTX and Word shown as PDFs)
Notifications	New course material, my file finished processing, flashcard reminders
AI Tutor & Chat	Answers appear as they're written; scope follows the workspace (my sources or course materials); web results are labeled

Instructor on mobile: lesson preparation

Workspaces	Create personal workspaces to prepare lessons
Sources & notes	Add files, photos, and links from the phone; write notes
Summaries	Summarize documents, chapters, videos, and links
Generate with AI	Quizzes and flashcards from selected sources

Instructor on mobile: classroom management

Classrooms & courses	Create and edit classrooms, each with one or more courses
Invitations	Share an invitation link or code; regenerate, revoke, set expiry
Co-instructors	Invite other instructors to a classroom
Students	Remove students from a classroom
Upload materials	All file types, YouTube and web links, subtitle files; uploads continue in the background
Processing	See status (processing, ready, failed), retry, or remove a material

Student on mobile: access

Login	Email or username and password, entered on the Keycloak login page in the phone's browser, so the first-login password change and forgotten-password reset just work
Joining	Invitation link that opens the app directly, or typing the code
Browsing	Classroom → course
Viewing	PDF viewer that jumps to a page (PPTX and Word shown as PDFs)
AI Tutor & Chat	Answers appear as they're written; scope follows the workspace (my sources or course materials); web results are labeled

Student on mobile: workspaces and study

Personal workspace	Create, rename, delete, favorite
Course workspace	Create from a joined course; the course materials are the sources, no uploads
My sources	In a personal workspace: add files and photos from the phone, links, or share from other apps
Quizzes	Choose sources, number of questions, and difficulty; all four question types with score and feedback
Flashcards	Swipe to flip and move; reminders to review

Thank you

AI-POWERED LEARNING ASSISTANT